An

INDUSTRIAL TRAINING REPORT

ON

MECHATRONIC INTERNSHIP AT TOBOTICS PVT. LTD.

Submitted in partial fulfilment of the award of Bachelor of Technology degree of Rajasthan Technical University, Kota

JAIPUR ENGINEERING COLLEGE
AND RESEARCH CENTRE

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CERTIFICATE

This is to certify that the industrial training done at “Tobotics Pvt. Ltd.” and this report is the bona fide work carried out by Parth, a student of B.Tech. (ECE) of Jaipur Engineering College and Research Centre, Jaipur, affiliated to Rajasthan Technical University, Kota, Rajasthan (India), during the academic year 2026–27, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (Electronics and Communication Engineering).

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I would like to express my sincere gratitude to Tobotics Pvt. Ltd. for providing me the opportunity to complete my industrial training in a practical and project-oriented environment. The training gave me exposure to electronics, Arduino programming, mechatronics, CAD modelling, 3D design and prototyping.

I am thankful to the mentors and senior members who guided me during the training, explained practical concepts and helped me understand how an idea can be converted into a working prototype. Their guidance was particularly useful while working on Arduino-based exercises and while developing the mechanical concepts for the Hackable Vault.

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Finally, I thank the fellow trainee students with whom I worked during the training. Working in a small group made it easier to discuss ideas, compare approaches and learn from practical problems encountered during development.

Parth

CONTENTS

Certificate II

Acknowledgement III

Contents IV

List of Figures VII

List of Tables VIII

List of Symbols / Abbreviations VIII

Abstract 1

1. Introduction 1

   1.1 Industrial Training Overview

   1.2 About Tobotics Pvt. Ltd.

   1.3 Training Objectives

   1.4 Scope of Training

   1.5 Training Environment

   1.6 Organization of Report

2. Training Technologies and Tools 4

   2.1 Arduino Platform

   2.2 Electronic Components and Interfacing

   2.3 Arduino Programming

   2.4 CAD and 3D Modelling

   2.5 3D Printing

   2.6 Simulation and Prototyping

3. Work Performed During Training 9

3.1 Arduino and Electronics Fundamentals

3.2 Debouncing Program

3.3 Gate-In / Gate-Out Project

3.4 Initial Art Thief Vault

3.5 First Vault Prototype and Simulation

3.6 CAD Modelling Exercises

3.7 Piston Cup Design and 3D Printing 17

3.8 Vault Front Panel and Hinge Studies 19

3.9 Design Iteration and Training Progress 21

4. Hackable Vault Project 24

4.1 Project Overview and Motivation 24

4.2 Concept Development

4.3 System Requirements

4.4 System Architecture

4.5 Hardware Design

4.6 Circuit Design and Simulation 29

4.7 Software and Game Logic

4.8 Timer and Difficulty Concept

4.9 Mechanical Design – Front Panel 32

4.10 Cam-Ring and Follower Mechanism

4.11 Door, Hinge and Locking Mechanism

4.12 Prototype Workflow and Current Status 35

5. Results and Discussion 37

5.1 Simulation and Programming Results

5.2 CAD and 3D Printing Results

5.3 Vault Design Results

5.4 Challenges and Design Modifications

5.5 Validation Plan

6. Learning Outcomes 41

6.1 Technical Learning

6.2 Mechanical-Electronic Integration

6.3 Problem Solving and Design Iteration

6.4 Professional and Practical Skills

7. Conclusion 43

7.1 Conclusion

7.2 Future Scope

Cost of Project 44

Appendices 45

References 51

LIST OF FIGURES

Figure 3.1 Debouncer circuit 10

Figure 3.2 Gate-In / Gate-Out project 11

Figure 3.3 Art Thief game reference 12

Figure 3.4 First Art Thief Vault prototype 14

Figure 3.5 Phone-holder CAD exercises 15

Figure 3.6 Piggy bank and component-box CAD exercises 16

Figure 3.7 Piston cup CAD model 18

Figure 3.8 Physical 3D-printed piston cup 19

Figure 3.9 Front-panel and vault enclosure CAD 20

Figure 3.10 Syblin concept and earlier mechanical mechanism 22

Figure 4.1 Hackable Vault reference 25

Figure 4.2 Wokwi electronic connection 29

Figure 4.3 Front-panel cam-ring mechanism 32

Figure 4.4 Internal mechanism concept 34

Figure 4.5 Overall vault enclosure CAD 35

Figure A.3 Front-panel mechanism CAD evidence 47

Figure A.4 Overall vault enclosure CAD evidence 47

Figure A.5 Piston cup CAD evidence 48

Figure A.6 Phone-holder CAD evidence 48

Figure A.7 Piggy bank CAD evidence 49

Figure A.8 Internship certificate 50

LIST OF TABLES

Table 2.1 Main electronic components and functions 5

Table 3.1 Training activities and skills developed 23

Table 4.1 Hackable Vault system requirements 28

Table 4.2 Prototype pin mapping 31

Table 5.1 Functional validation plan 41

Table C.1 Approximate project cost 44

LIST OF SYMBOLS / ABBREVIATIONS

Term	Meaning
Arduino	Microcontroller development platform used for the training prototypes
CAD	Computer-Aided Design
ECE	Electronics and Communication Engineering
I2C	Inter-Integrated Circuit communication interface
LED	Light Emitting Diode
OLED	Organic Light Emitting Diode display
PWM	Pulse Width Modulation
USB	Universal Serial Bus
Wokwi	Online electronics simulation platform used for the prototype
STEM	Science, Technology, Engineering and Mathematics

ABSTRACT

Industrial training provides an opportunity to connect classroom concepts with practical engineering work. This report presents the work completed during a 45-day industrial training at Tobotics Pvt. Ltd., a college-incubated startup where the training focused on Arduino-based electronics, embedded programming, mechatronics, CAD modelling, 3D design and prototyping.

The training progressed from basic Arduino programming and electronic interfacing to application-oriented exercises and mechanical design. Practical work included a switch debouncing program, a Gate-In/Gate-Out occupancy-control concept, an initial Art Thief Vault prototype, CAD modelling exercises, and the design and 3D printing of a piston cup model.

The major project developed during the training was the Hackable Vault, an interactive security-vault concept combining keypad input, display feedback, sound and light indications, servo-controlled movement, and a mechanically designed vault enclosure. The project evolved through several design iterations as mechanical complexity and space constraints were evaluated. The final project reached a substantially developed prototype/design stage, while complete software and full mechanical integration remained areas for further work.

CHAPTER 1

INTRODUCTION

Industrial training is an important part of engineering education because it gives students an opportunity to work with practical tools, components and design processes. Classroom subjects introduce the principles of electronics, programming and engineering design, but practical training shows how these principles behave when they are combined into an actual system. The training described in this report was undertaken at Tobotics Pvt. Ltd. as part of the mandatory industrial training requirement of the B.Tech. Electronics and Communication Engineering programme.

The training had a practical and project-oriented character. Rather than remaining limited to theoretical discussion, the work involved writing Arduino programs, connecting electronic modules, testing behaviour through simulation, designing mechanical parts using CAD software and converting selected digital designs into physical objects through 3D printing. This combination was particularly useful because the final project required both electronic control and mechanical movement.

The report therefore treats the internship as a complete learning process. The project is included as the major application of the skills developed during the training, while smaller exercises are also documented because they show how the learning progressed from basic concepts toward a larger integrated design.

1.2 About Tobotics Pvt. Ltd.

Tobotics Pvt. Ltd. is the organization where the industrial training was completed. According to the training context, the company is a startup associated with the college incubation environment. The training therefore took place in an environment that was closely connected to the academic ecosystem while still being oriented toward practical product and prototype development.

The work environment included other trainee students and mentors who guided the learning activities. The organization was described as working around mechatronics, STEM-oriented projects and interactive hardware concepts. The training consequently provided exposure to both electronic and mechanical aspects of product development rather than focusing on a single ECE topic.

An important feature of the training was the gradual movement from guided exercises to independent design decisions. Initial tasks were comparatively small and focused on learning a particular programming or design feature. Later work required the trainee to decide how a mechanism should operate, how electronic inputs should control outputs and how a physical prototype could be made practical within mechanical constraints.

1.3 Training Objectives1.4 Scope of Training

The primary objective of the industrial training was to obtain practical exposure to electronics and mechatronics through hands-on work. The training was intended to improve the ability to work with Arduino boards, electronic components, programming logic, CAD software and physical prototypes.

Specific learning objectives included understanding basic Arduino programming, interfacing input and output devices, controlling actuators, testing programs through simulation, creating mechanical models in CAD software and understanding the transition from a digital model to a physical 3D-printed part. Another objective was to learn how a larger project can be developed through smaller experiments and repeated design changes.

The scope of the training covered two major phases. The first phase concentrated mainly on Arduino and electronic work. The second phase moved toward 3D modelling, CAD and 3D printing. The Hackable Vault project combined these areas by requiring an electronic game/control system together with a mechanical enclosure and moving components.

1.5 Training Environment1.6 Organization of Report

The training was carried out in a small, practical group environment with other trainee students and mentors. The work was not limited to demonstrations: programs were tested, circuits were changed during prototyping, and CAD ideas were revised when a mechanism became difficult to build. The close interaction with mentors and senior members made it possible to discuss design decisions while the work was still being developed.

This report is organized to follow the industrial-training requirements while keeping the technical story connected. Chapter 1 introduces the training and organization. Chapter 2 explains the main technologies and tools used. Chapter 3 records the work performed during training, including the smaller exercises and design activities. Chapter 4 gives a detailed account of the Hackable Vault project. Chapter 5 discusses results, testing, challenges and design changes. Chapter 6 summarizes the learning outcomes, and Chapter 7 concludes the report and presents future scope. Cost, appendices and references are provided after the chapters.

The structure also separates the learning process from the final project. This is important because several of the skills used in the final vault were first developed through smaller exercises.

CHAPTER 2

TRAINING TECHNOLOGIES AND TOOLS

The training introduced a combination of electronic control, programming and mechanical design. The technologies were not learned independently; they were gradually connected through practical exercises. Arduino was used as the main controller platform for electronic activities, while CAD software was used for mechanical modelling and 3D printing was used to create selected physical parts.

The overall technology stack can be viewed as a simple pipeline: an idea is converted into a program or CAD model, tested through simulation or inspection, modified when necessary, and then implemented physically when the design is suitable. This workflow was later applied to the Hackable Vault, where software logic and mechanical design had to be considered together.

The following sections describe the major tools and concepts that formed the technical foundation of the training.

2.2 Electronic Components and Interfacing

The electronic work used common modules and components suitable for Arduino-based prototyping. The project context includes Arduino boards, keypads, displays, servo motors, LEDs, a buzzer, resistors, breadboards and connecting wires. These components allowed input, processing and output functions to be demonstrated without requiring a permanent PCB.

An Arduino board served as the controller. Input devices such as a keypad or switches supplied information to the program. Output devices such as LEDs, a display, buzzer and servo motors provided visible, audible or physical responses. Breadboards and jumper connections were useful during prototyping because the circuit could be changed without permanently soldering the components.

The use of several modules together also introduced an important practical consideration: every component needs both an electrical connection and a software control strategy. A component

can be physically connected and still fail to perform the intended task if the program does not read or drive the correct pin.

Component	Use in training/project
Arduino Uno/Nano	Controller for prototypes and intended final build
4×4 keypad	Numerical user input
LCD/OLED display	Level, time and feedback information
Servo motor	Physical actuation
LEDs and buzzer	Visual and audible feedback
Breadboard/wires	Rapid prototyping and circuit changes

Table 2.1: Main electronic components and their functions.

2.3 Arduino Programming

Arduino programming formed the software foundation of the training. The programs were developed around the controller's setup and loop structure, with functions used to divide larger behaviour into smaller tasks. Conditional statements were used to make decisions based on user input, while variables stored states such as counters, guesses and level information.

One of the early exercises focused on switch debouncing. Mechanical switches can produce multiple rapid transitions when pressed, even though the user intends a single action. The exercise helped demonstrate why input signals sometimes need to be processed rather than used directly.

The later vault code used similar programming principles at a larger scale. The program reads a keypad input, builds a password guess, compares it with a generated value, provides HIGH/LOW feedback and changes the system state after a correct answer. The uploaded prototype code also demonstrates the use of functions such as startLevel() and winGame() to keep the program organized.

2.4 Keypad and Display Interfacing

A keypad provides a convenient way to enter numerical information into an Arduino system. In the uploaded vault prototype, a 4×4 keypad is configured with four rows and four columns. The program receives a character when a key is pressed and then distinguishes between numerical keys and control keys such as # and *.

The display provides immediate feedback to the user. In the prototype code, the display shows the current level, prompts the user to enter a code, reports HIGH or LOW after an incorrect guess and displays CORRECT or VAULT OPEN at important stages. This makes the system interactive rather than simply producing an invisible change in a variable.

Interfacing an input and display together also demonstrates the importance of state management. The program has to know whether the user is entering a number, submitting a guess, clearing the input, progressing to the next level or waiting for a reset.

2.5 Servo Motor Interfacing

Servo motors were used as the main actuators in the prototype work. A servo can be commanded to move to a specified angular position, making it suitable for small mechanical movements. In the uploaded prototype code, four servos are attached to different Arduino pins and initialized to a closed position.

In the first vault prototype, a successful password causes the servo corresponding to the current level to move to a different angle. This provided a simple way to translate a correct software decision into a physical action. The approach was useful as an early proof of concept because each successful level could be associated with one physical movement.

The later Hackable Vault concept reduced the mechanism to two servos. One servo is intended to rotate a ring mechanism that turns four decorative front knobs, while the second servo is intended to operate the actual vault door or locking mechanism. This design uses mechanical transmission to reduce the number of independently controlled actuators.

2.6 CAD and 3D Modelling

The later part of the training concentrated on CAD and 3D modelling. CAD made it possible to design components digitally before making them physically. The work included simple objects such as phone holders and a piggy bank, followed by more application-specific components for the vault.

CAD was particularly useful for the Hackable Vault because the mechanism contains moving parts whose positions must be considered together. The front panel includes four knobs and a ring-based mechanism. A separate hinges test model was used to explore the movement of a rectangular vault-like door, a locking arrangement and a mechanism intended to push the panel outward after unlocking.

These exercises also demonstrated that mechanical design is iterative. A mechanism that looks reasonable in a sketch may need changes after considering available space, movement range, attachment points and the interaction between parts.

2.7 3D Printing

3D printing provided the physical link between CAD design and a tangible object. During the training, a piston cup model was designed and 3D printed. The physical print served as evidence that a digital model could be prepared for fabrication and converted into an actual object.

The piston cup exercise was useful because it was a relatively compact object while still demonstrating the complete CAD-to-print workflow. It also provided an opportunity to observe the difference between a model on a screen and a real part, including its size, surface and physical handling.

For the vault project, the same general approach is relevant. Parts must not only look correct in CAD; they also need suitable dimensions, clearances and attachment points for assembly. The training therefore helped connect visual design with practical manufacturing considerations.

2.8 Simulation and Prototyping

Simulation was used during the electronic development of the vault. The Wokwi project contains an Arduino Uno, keypad, display, servo motors, LEDs, buzzer, resistors and breadboard connections. The simulation made it possible to check the basic relationship between the program and connected components before relying on a complete physical assembly.

The value of simulation was especially clear during the early vault stage. A simple prototype could be changed quickly by modifying connections or code. This reduced the risk of spending physical components and time on a design that had not yet been logically checked.

Prototyping also encouraged an iterative approach. The first design was not treated as permanent. The project moved from a simple Art Thief Vault concept to a more ambitious monster-based toy concept and then to the Hackable Vault. Each change was driven by the practical difficulty of the previous approach.

CHAPTER 3

WORK PERFORMED DURING TRAINING

The work performed during the training can be divided into a sequence of practical exercises and project-development activities. The early activities focused on understanding Arduino programming and electronic inputs and outputs. Later activities involved CAD, mechanical design and 3D printing. The major project developed across these phases.

This chapter records the work as a progression rather than as isolated assignments. Each activity contributed a skill that was later used in the larger project. The smaller programming exercises improved understanding of input processing and control logic, while the CAD exercises developed the ability to create and inspect physical mechanisms.

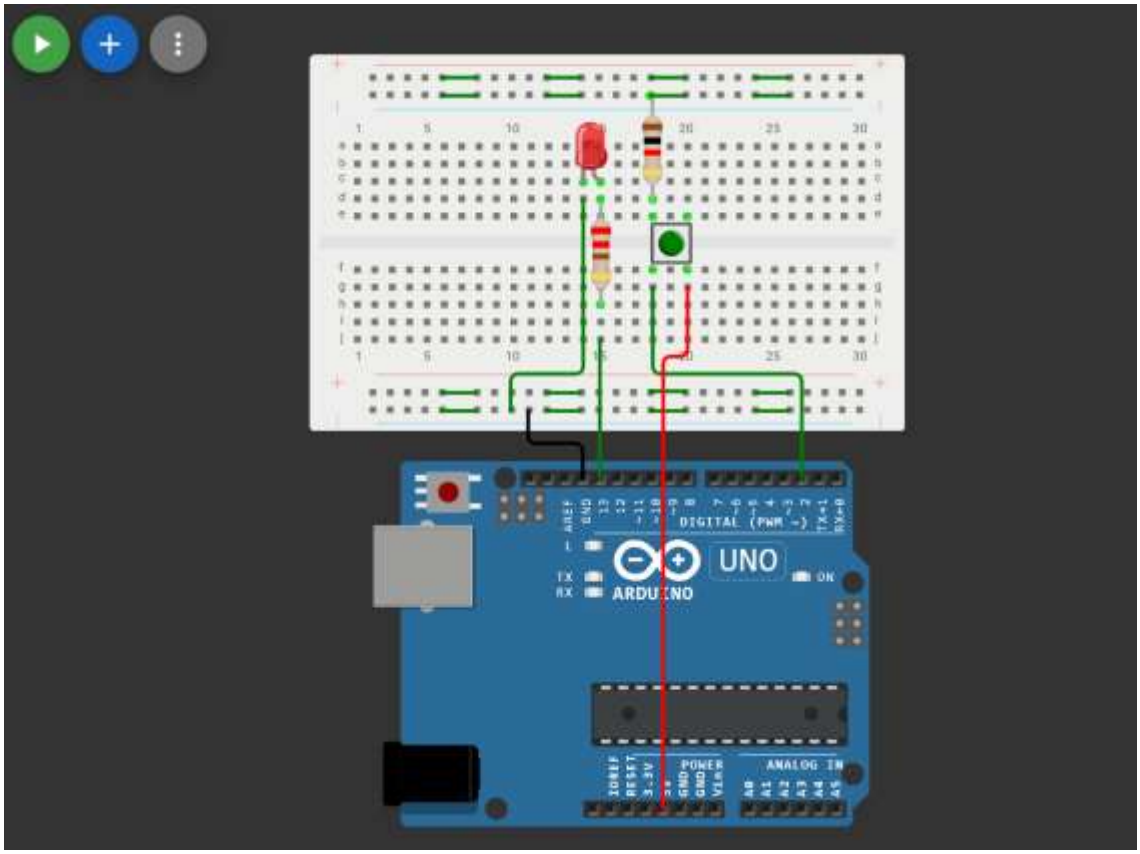


Figure 3.1: Debouncer circuit used during the initial Arduino exercise.

3.2 Debouncing Program

One of the early programming exercises involved switch debouncing. The purpose was to understand the behaviour of a mechanical switch when it is pressed or released. Instead of assuming that one physical press always creates one perfectly clean digital transition, the program considers the short unstable period around the switching event.

The exercise was important because it introduced a practical difference between ideal circuit diagrams and real hardware. A controller reads electrical signals, and those signals can contain transitions that need to be interpreted correctly by software.

The learning from this exercise was later relevant to interactive projects. A keypad or button-based system should respond to deliberate user actions rather than accidental repeated transitions. Debouncing therefore provided a basic example of how software can improve the reliability of a physical interface.

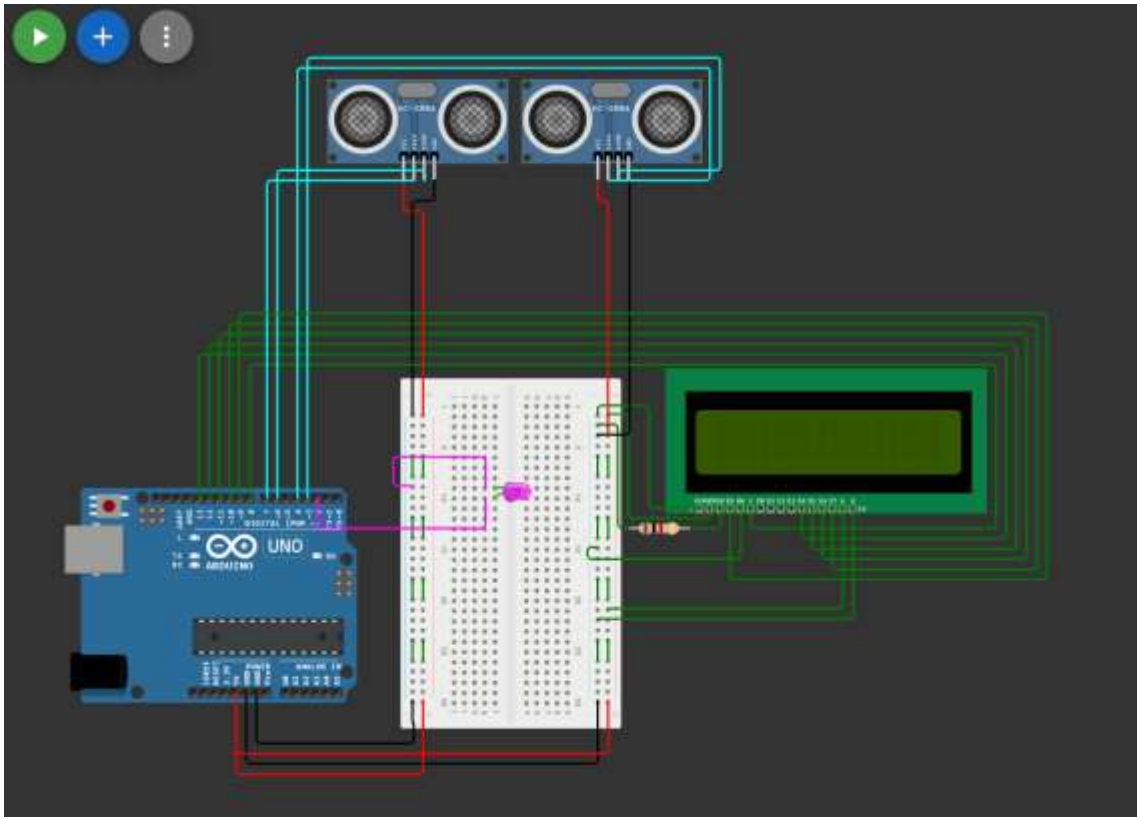


Figure 3.2: Gate-In / Gate-Out project used to demonstrate occupancy-based control.

3.3 Gate-In / Gate-Out Project

The Gate-In/Gate-Out project was a small application developed using an Arduino. The concept used two gates: an entry gate and an exit gate. When a person enters through the entry side, the system increases the room occupancy count. When a person leaves through the exit side, the count decreases.

The purpose of the project was to connect sensor/input events with a stored count and then use the count to control an output condition. When the system determines that people are present, the connected lighting and fan loads are intended to remain active. When the room becomes empty, the power can be switched off to reduce unnecessary consumption.

The project introduced a simple form of automation: the controller does not merely display a value but uses that value to decide what the system should do.

3.4 Gate-In / Gate-Out Working

The logic of the Gate-In/Gate-Out system can be represented as a sequence. First, the controller waits for an input event at either gate. An entry event increases the occupancy value by one, while an exit event reduces it by one. The updated count is then used to determine whether the room should be considered occupied.

When the occupancy is greater than zero, the system can keep the required electrical loads active. When the occupancy reaches zero, the loads can be turned off. The project therefore demonstrates a basic closed-loop decision process in which input information changes an internal state and the state controls outputs.

This exercise was useful preparation for the Hackable Vault. The vault also follows the same broad structure: user input is received, the controller changes an internal state, and the state produces a physical or visual response.

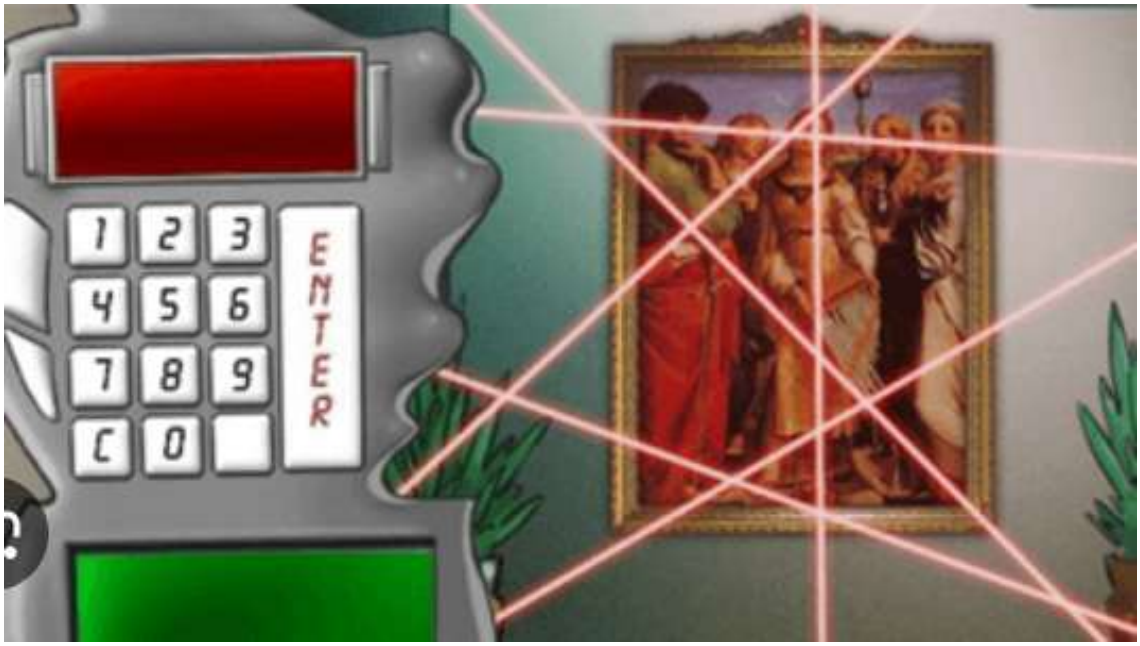


Figure 3.3: Reference for the original Art Thief password-guessing concept.

3.5 Initial Art Thief Vault Concept

The first personal project concept was inspired by an old Flash game involving an art thief. In the game concept, the user enters a numerical guess and receives a HIGH or LOW indication that helps identify the hidden password. Correct progress removes barriers protecting the artwork.

For the initial physical prototype, the idea was translated into an electronic vault. A keypad was used for entering guesses, a display was used for feedback, and servos represented the physical barriers. The concept was intentionally simple so that the basic game interaction could be tested before attempting a complex mechanical enclosure.

The project was useful because it connected several training topics in one small system: keypad input, password logic, display feedback, servo movement and audible/visual indication.

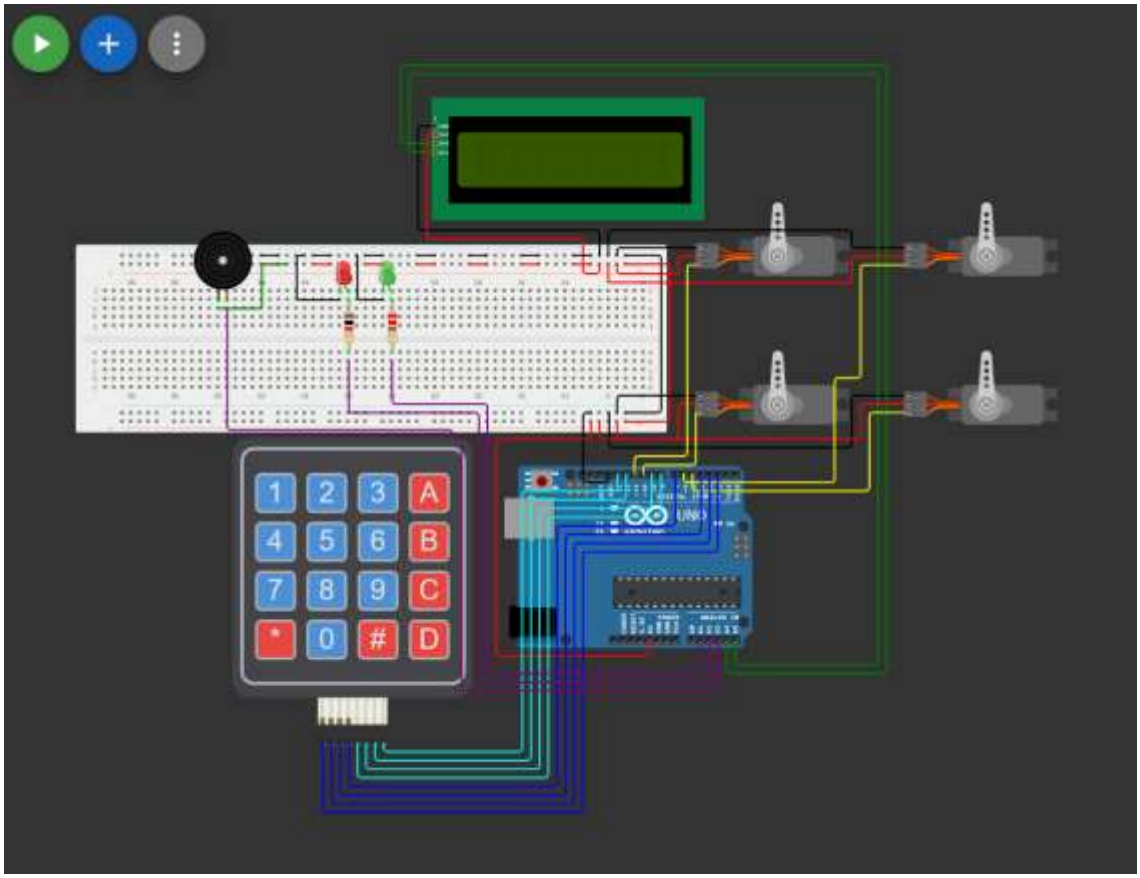


Figure 3.4: First Art Thief Vault prototype concept with keypad, display and servo outputs.

3.5 First Vault Prototype and Simulation

The first vault prototype was implemented in simulation with an Arduino, keypad, display, LEDs, buzzer and four servo outputs. The uploaded circuit configuration shows the Arduino Uno connected to the keypad, display and four servo motors, with separate LED and buzzer connections.

The corresponding program generates a two-digit secret password, accepts a two-digit user input and compares the guess with the generated value. An incorrect guess produces HIGH or LOW feedback, while a correct guess moves the servo associated with the current level and advances the game. After four successful levels, the program enters the vault-open state.

This first version served as a functional proof of the electronic game logic even though it was not the final physical design.

Prototype element	Function
Arduino Uno	Controller
4×4 keypad	Password input
16×2 I2C LCD	User feedback
Four servos	Stage/barrier movement
Green / red LEDs	Correct / incorrect indication
Buzzer	Audible feedback



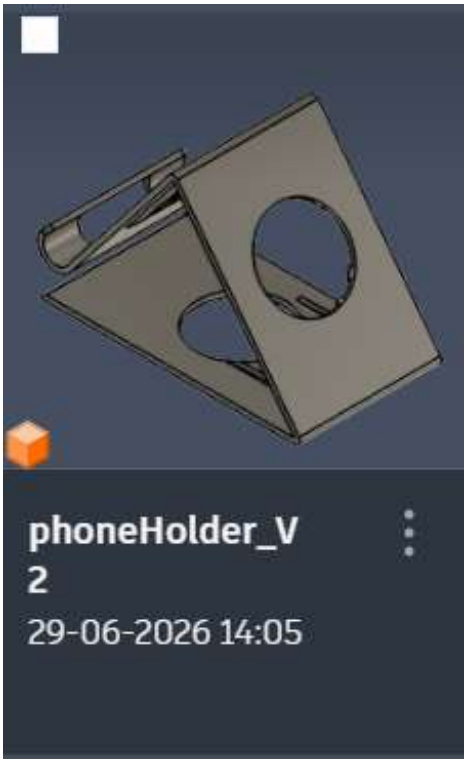


Figure 3.5: Two phone-holder CAD exercises produced during training.

3.6 CAD Modelling Exercises

The CAD phase began with simpler modelling exercises. These tasks were useful for learning how sketches, dimensions, extrusions and other modelling operations can be combined to form a physical object.

One of the first assignments involved designing a phone holder or stand. Two versions were made as part of the learning process. The task encouraged the trainee to use imagination and determine how a simple functional object could be represented as a CAD model.

These simple models created a foundation for later work. The vault required much more attention to movement and assembly, so understanding how individual solids and components are constructed was necessary before attempting the larger mechanism.

3.6.1 Phone Holder, Piggy Bank and Component Box

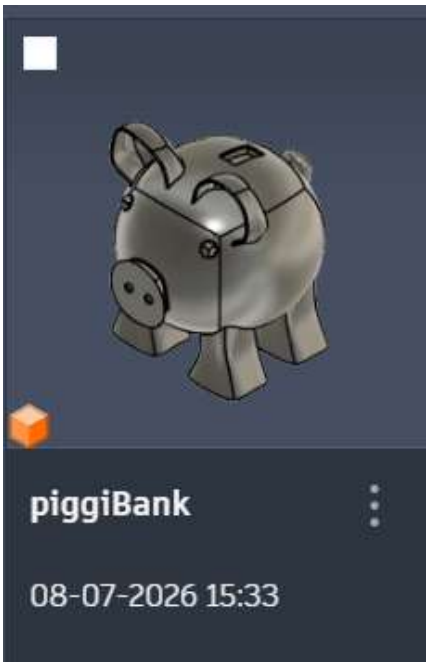


Figure 3.6: Piggy bank CAD exercise.



Figure 3.6(a): Electronic-component-box CAD exercise.

The phone holder exercise focused on creating a useful object from a relatively simple geometric arrangement. The work helped develop confidence with basic CAD construction and the idea that a model should have a practical purpose rather than only being an abstract shape.

A piggy bank model was also designed to explore more organic forms. This exercise was useful because the geometry was less dependent on simple rectangular features and required a different way of thinking about shape.

Together, these exercises broadened the modelling experience. The phone holder represented a simple functional product, while the piggy bank provided an opportunity to experiment with curved or organic geometry. The lessons from both were later useful when considering the external form and mechanical components of the vault.



Figure 3.7: Piston cup CAD model.



Figure 3.8: Physical 3D-printed piston cup produced during the training.

3.7 Piston Cup Design and 3D Printing

A piston cup model was designed during the CAD phase and was selected as one of the objects for physical printing. The design was based on the recognizable piston-cup form associated with the Cars movie universe, but the training value came from the modelling and fabrication process rather than the reference itself.

The model provided a practical opportunity to examine how a designed object would translate into a printable geometry. It also served as a personal milestone during the training because it resulted in a physical object rather than only a digital file.

The CAD model demonstrated the ability to move from an idea to a structured 3D representation and then prepare the design for physical fabrication.

The piston cup was subsequently 3D printed during the training. The printed object is a useful piece of physical evidence because it shows the complete path from CAD design to manufacturing.

During a digital design exercise, it is easy to judge success only from the appearance of a model on screen. The physical print provides a different perspective: dimensions become tangible, the part can be handled, and design choices can be evaluated against the real object.

The piston cup therefore represents an important transition in the training from purely digital work toward physical prototyping. The same transition is relevant to the Hackable Vault, where the final goal is not only to create a CAD assembly but to build a mechanism that can physically move and interact with the user.

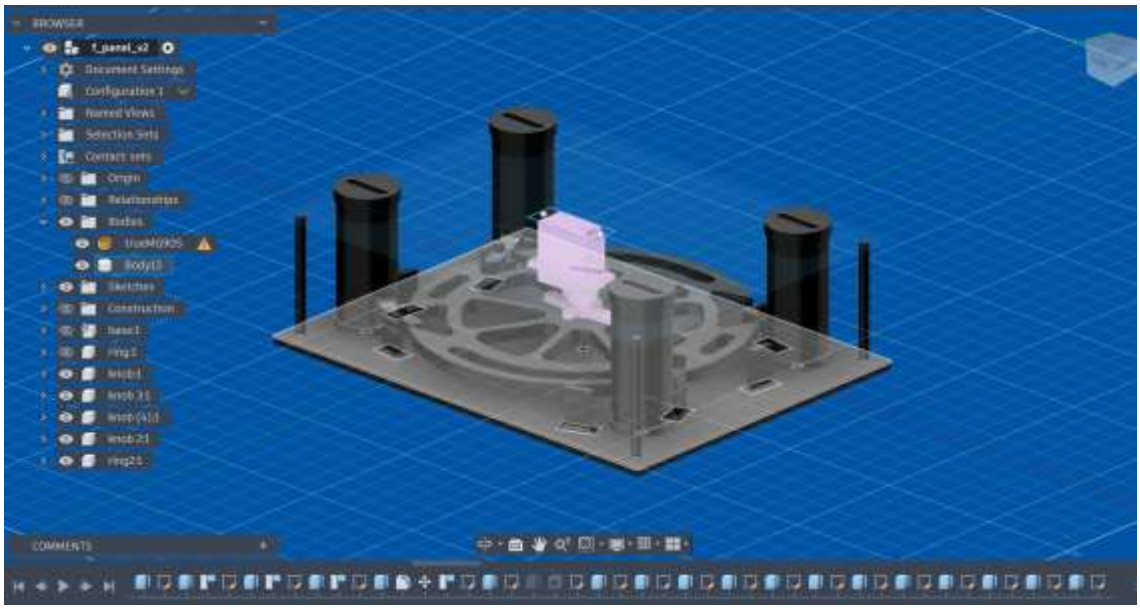


Figure 3.9: Front-panel CAD mechanism for the four rotating knobs.

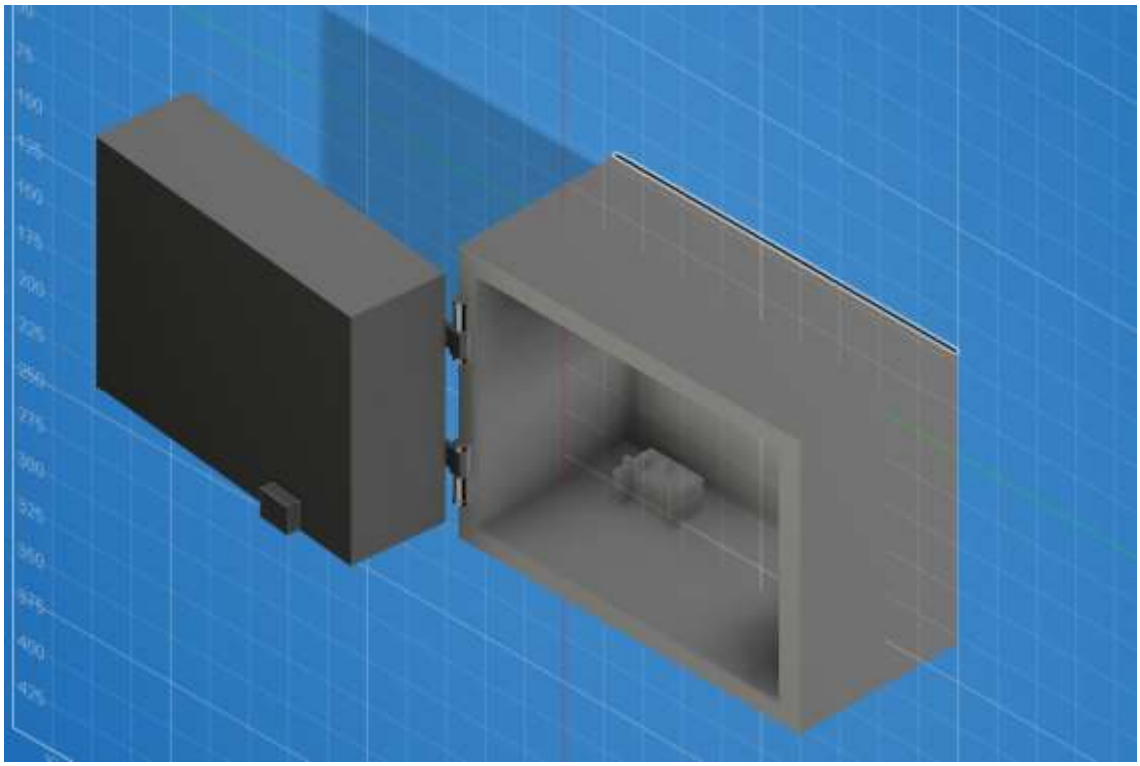


Figure 3.9(a): Overall vault enclosure CAD study.

3.8 Vault Front Panel and Hinge Studies

The front panel of the Hackable Vault was developed as a dedicated CAD assembly. The design includes four rotating knobs arranged around the front panel and a central ring mechanism intended to rotate them together.

The file referred to during development as F_panel_V2 contains the mechanism for the knob movement and the servo-related arrangement. The basic idea is to place one servo at the centre of a large ring. The ring contains ramps or raised profiles that interact with followers connected to the knobs.

The advantage of this approach is that four visible movements can be produced from one actuator. The knobs are primarily an animation and user-feedback mechanism; their purpose is to make the unlocking process physically visible rather than to act as four independent electronic locks.

A separate CAD model referred to as the hinges test was used to study the physical vault enclosure. The model consisted of a rectangular vault-like structure with a hinged front panel. This allowed the door movement to be considered independently from the more complex front-knob mechanism.

The design also explored a locking mechanism driven by a servo. Another feature was intended to maintain pressure on the front panel while the lock was engaged. When the locking element moved away from the panel, the stored pressure would push the door outward by approximately a few centimetres, making the opening action visually obvious.

This model was important because it isolated the mechanical problem of opening and locking the vault. It also showed why mechanical design required repeated testing and refinement.



Figure 3.10: Syblin concept considered during the intermediate project stage.

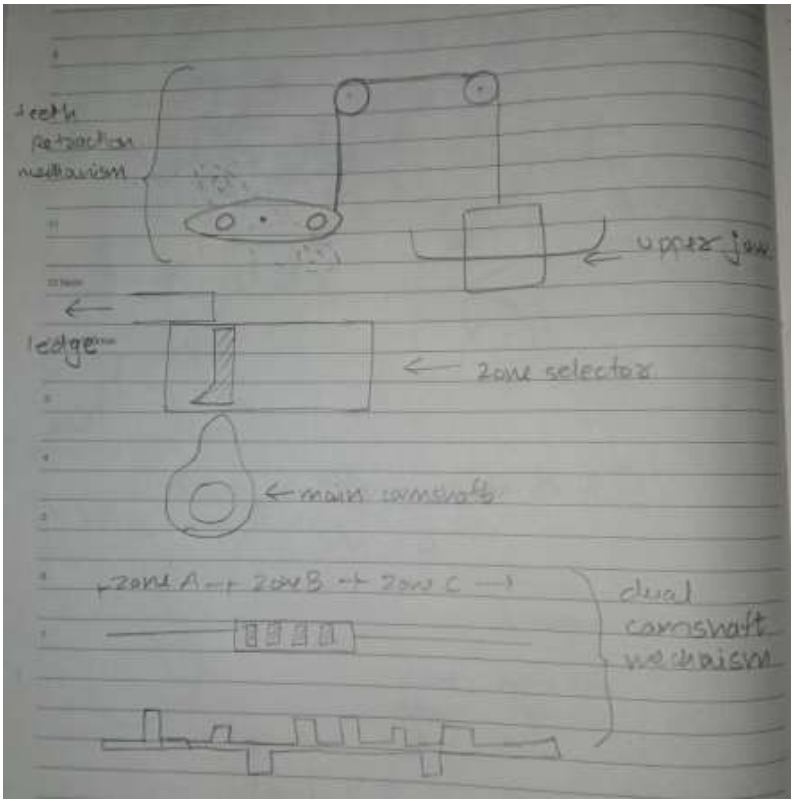


Figure 3.10(a): Earlier internal mechanism concept considered for the Syblin version.

Activity	Main skill developed
Debouncing	Reliable handling of mechanical input
Gate-In / Gate-Out	State-based counting and automation
Art Thief Vault	Keypad, display, password and servo integration
CAD exercises	Sketching, modelling and assembly thinking
Piston cup	CAD-to-3D-print workflow
Hackable Vault	Integration of electronics, software and mechanism

Table 3.1: Training activities and corresponding skills developed.

3.9 Design Iteration and Training Progress

The personal project changed substantially during development. The original idea was a simple Art Thief Vault. It was then expanded into a monster character called Syblin, a combination of goblin and cyclops, with a screen-like eye and retracting teeth.

The teeth concept introduced a significant mechanical challenge. A separate servo for every tooth would increase the number of actuators and occupy considerable space. A mechanical alternative using a camshaft and zone-selection mechanism was considered to reduce the number of servos while introducing different tooth sequences.

After evaluating the complexity of the mechanism, the project was redirected toward the Hackable Vault. This change reduced the mechanical complexity while preserving the central idea of password guessing followed by a satisfying physical unlocking action.

The overall training progressed from basic exercises toward an integrated prototype. The early programming work established familiarity with Arduino logic and physical inputs. The Gate-In/Gate-Out project then demonstrated how stored state could control real-world outputs.

The first vault prototype combined these ideas into an interactive game. The later CAD phase added the mechanical design skills required to create a physical enclosure and moving parts. The piston cup provided a completed CAD-to-3D-print example, while the vault front panel and hinges test represented more advanced mechanical experimentation.

By the end of the training, the Hackable Vault had reached a substantially developed design stage. The electronic connections had been planned and simulated, the main front-panel mechanism had been modelled, and a separate door/hinge mechanism had been explored. Full code and complete mechanical integration remained unfinished.

CHAPTER 4

HACKABLE VAULT PROJECT



Figure 4.1: Reference for the Hackable Vault form and user-interface arrangement.

4.1 Project Overview and Motivation

The Hackable Vault is the major personal project developed during the training. It is an interactive security-vault concept in which the user attempts to unlock a physical vault by entering a sequence of correct numerical guesses.

The project combines three areas developed during the internship: embedded programming, electronic interfacing and mechanical CAD design. The electronic system handles user input,

timing and feedback, while the mechanical system converts servo movement into visible physical motion.

The project was designed as an interactive prototype rather than as a commercial security product. The intended experience is simple: the user enters guesses, receives HIGH/LOW feedback and eventually sees a physical unlocking sequence. The combination of display feedback, sound, LEDs and servo-driven movement was chosen to make the result visible and satisfying.

The motivation for the project came from the idea of turning a password-guessing game into a physical object. Instead of displaying the result only on a computer screen, the user interacts with a keypad attached to the vault and receives feedback from displays, LEDs and a buzzer.

The intended interaction is simple. The user begins a level, enters a numerical guess and submits it. The system indicates whether the guess is HIGH or LOW relative to the hidden value. The user uses this information to make another guess. A correct answer triggers a physical action and moves the system toward the next level.

The design also aims to make the final successful action visually satisfying. The four front knobs are intended to rotate before the door mechanism operates, providing a physical indication that the vault is being unlocked.

4.2 Concept Development

The project went through several concepts before reaching the current Hackable Vault design. The first inspiration was the Art Thief Flash game, where a numerical guessing mechanic was connected to the removal of barriers protecting an artwork.

The first physical interpretation used a cage or vault-like structure with servo-controlled barriers. The idea was then expanded into Syblin, a goblin-cyclops monster with retractable

teeth. The tooth design led to a proposed dual-camshaft mechanism so that many teeth could be controlled using only a small number of servos.

Because the dual-camshaft and tooth-retraction mechanism introduced substantial mechanical complexity, the concept was simplified. The Hackable Vault retained the password guessing, level progression and physical reward but replaced the large number of moving teeth with a four-knob front panel and a separate door mechanism.

4.3 System Requirements

The electronic requirements for the Hackable Vault include a microcontroller, a numerical input device, displays, visual indicators, an audible indicator and servo actuators. The selected hardware context includes an Arduino Nano for the intended build, two OLED screens, an LCD used during earlier prototyping, LEDs, a buzzer and associated wiring and connectors.

The mechanical requirements include a vault-like enclosure, a hinged front door, four decorative rotating knobs, a central ring mechanism, followers or levers for transferring ring movement to the knobs, and a locking/opening mechanism controlled by a second servo.

From a software perspective, the system needs password generation, user input handling, comparison logic, HIGH/LOW feedback, level progression, a timer, reset behaviour and a final vault-open sequence. Some of these functions were already demonstrated in the earlier prototype, while others were identified as remaining development tasks.

Area	Requirement
Input	4×4 keypad for password entry
Controller	Arduino Nano for intended build
Displays	Two OLED screens for time and game feedback
Audio/visual	Buzzer and LEDs for status indication
Actuation	Two servos: front-knob animation and door/lock

Mechanical	Vault enclosure, hinged door, four knobs and cam-ring

Table 4.1: Functional requirements of the Hackable Vault.

4.4 System Architecture

The system can be divided into four major blocks: user input, controller logic, feedback and physical actuation. The keypad is the main user-input device. The Arduino processes the entered value and compares it with the current password.

Feedback is provided through the displays, LEDs and buzzer. One display is intended to communicate the remaining time, while another can present the HIGH/LOW result or other game information. The buzzer provides different audible indications for wrong answers, correct answers and successful completion.

The actuator block contains two servos. The first servo controls the visible four-knob mechanism through the cam-ring arrangement. The second servo is intended to operate the actual door or lock mechanism. This architecture allows a relatively small electronic system to control a more complex physical interaction.

Block	Role
User / Keypad	Accepts password guesses
Arduino	Runs game logic and controls outputs
OLED / Display	Shows time and game information
LEDs	Provide visual status
Buzzer	Provides audible feedback
Servo 1	Drives four-knob front-panel animation
Servo 2	Operates door / locking mechanism

4.5 Hardware Design

The intended controller for the final Hackable Vault is an Arduino Nano. The electronic parts acquired for the project include displays, OLED modules, LEDs, a keypad, a buzzer and other supporting components. The exact final enclosure and component placement are still subject to mechanical refinement.

The keypad is placed on a separate attached hacking-device section. This makes the interaction visually distinct from the vault itself: the user operates the hacking interface while observing the vault as the physical target.

The front panel contains four knobs. Unlike a conventional four-lock vault, these knobs are primarily for visual animation. They are driven together by a single servo through the ring mechanism, allowing the design to produce a coordinated physical movement without dedicating one servo to each knob.

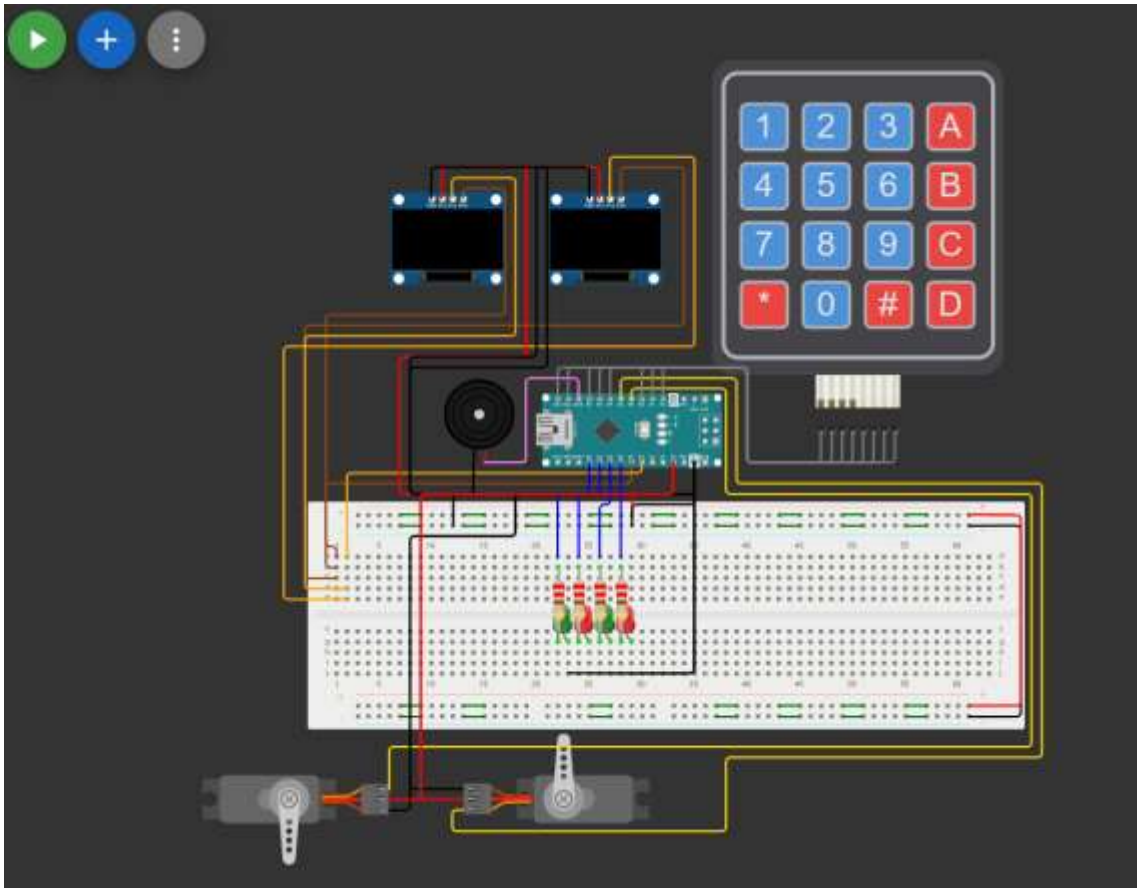


Figure 4.2: Wokwi electronic connection for the first vault prototype.

4.6 Circuit Design and Simulation

The electronic prototype was simulated in Wokwi. The uploaded circuit contains an Arduino Uno, a 4×4 membrane keypad, a 16×2 I2C LCD, four servo motors, red and green LEDs, a buzzer, resistors and breadboard power rails. The prototype demonstrates the basic connections used to validate the early game logic.

The keypad rows are connected to digital pins 2, 3, 4 and 7, while the columns use pins 8, 9, 12 and 13. The LCD uses the I2C interface through A4 and A5. In the prototype, the servo signals use pins 11, 10, 6 and 5, while the green LED, red LED and buzzer use A1, A2 and A3 respectively.

This simulation represents the earlier four-servo prototype rather than the final two-servo Hackable Vault hardware. It is therefore included as a development-stage circuit and not as a claim that the final hardware was fully assembled.

Device / signal	**Prototype connection**
Keypad rows	D2, D3, D4, D7
Keypad columns	D8, D9, D12, D13
I2C LCD SDA	A4
I2C LCD SCL	A5
Servo 1	D11
Servo 2	D10
Servo 3	D6
Servo 4	D5
Green LED	A1
Red LED	A2
Buzzer	A3

Prototype signal	Pin

Keypad rows	D2, D3, D4, D7
Keypad columns	D8, D9, D12, D13
I2C LCD	SDA A4, SCL A5
Servo outputs	D11, D10, D6, D5
Green LED	A1
Red LED	A2
Buzzer	A3

Table 4.2: Prototype pin mapping taken from the uploaded Arduino source and Wokwi configuration.

4.7 Software and Game Logic

The prototype software is organized around a small number of states and functions. The setup function initializes the display, attaches the servos, configures the LEDs and buzzer and starts the first level. The startLevel function generates a new password and resets the input state.

The main loop reads a keypad character. Numerical keys are appended to the current input until the allowed length is reached. The # key submits the guess, while * clears the current input. This makes the keypad operate as both a numerical input device and a control interface.

When the guess matches the generated password, the current servo moves, the green LED and buzzer are activated and the level counter increases. If the guess is incorrect, the program clears the input, displays HIGH or LOW and activates the red LED and buzzer for a short period.

The uploaded prototype uses a generated two-digit secret value. A new value is selected when a level begins. The user enters a numerical guess and submits it. If the guess is higher than the secret value, the display reports HIGH; if it is lower, the display reports LOW.

After a correct answer, the program advances the current level. The prototype contains four servo-controlled stages and therefore progresses through four successful levels before entering the win state. The level-specific servo movement provides a simple demonstration of how a software state can be mapped to a physical event.

The final Hackable Vault concept is intended to preserve the same basic game structure while expanding it with a timer, increasing difficulty and a two-servo physical unlocking sequence.

4.8 Timer and Difficulty Concept

A timer was identified as an additional feature for the Hackable Vault. The purpose is to turn the password guessing task into a race against time. A countdown can be displayed on one of the OLED screens, and the level should restart if the allotted time expires.

Difficulty can also increase between levels. The exact final implementation was not completed in the training period, but the design idea was to make later levels harder than earlier ones. One possible approach considered during development was to increase the numerical range used for the hidden password as the level increases.

The timer and difficulty functions are therefore treated in this report as planned project features rather than completed results. They form part of the future software work required to turn the current design into a complete game.

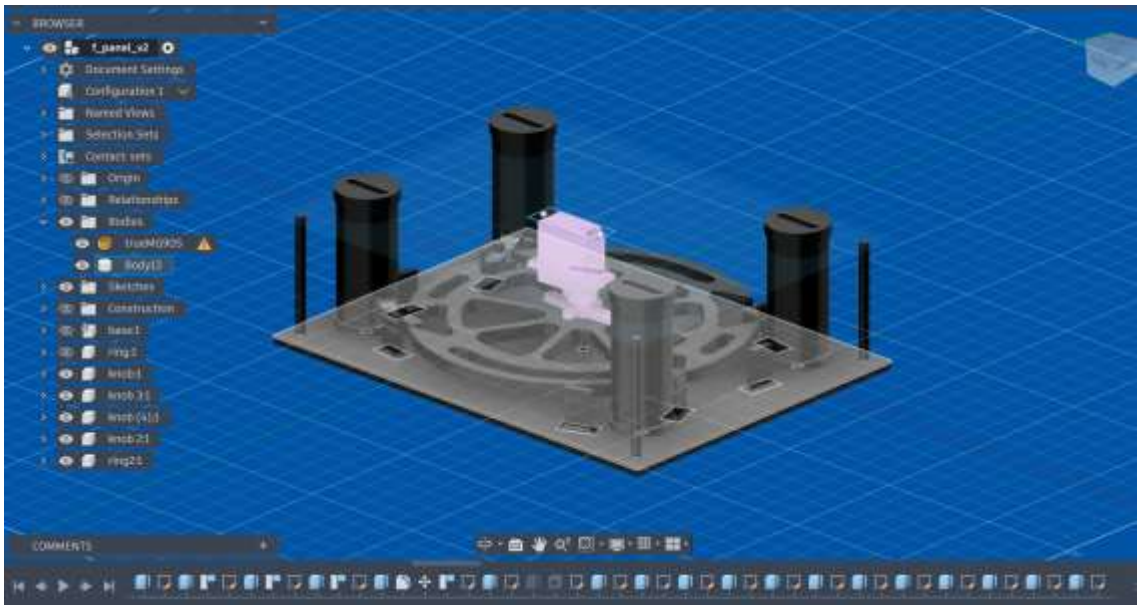


Figure 4.3: Front-panel cam-ring mechanism used to coordinate the four visible knobs.

4.9 Mechanical Design – Front Panel

The front panel is one of the most developed mechanical sections of the Hackable Vault. It contains four knobs positioned around the door and a large central ring connected to a servo. The ring contains raised ramps or profiles that interact with extensions from the knobs.

When the servo rotates the ring through a selected angle, the ramps are intended to move the followers and rotate the four knobs. The objective is for the four knobs to move together by approximately 90 degrees as part of the unlocking animation.

This approach is mechanically different from using one servo per knob. It reduces the number of actuators and makes the front panel behave as one coordinated mechanism.

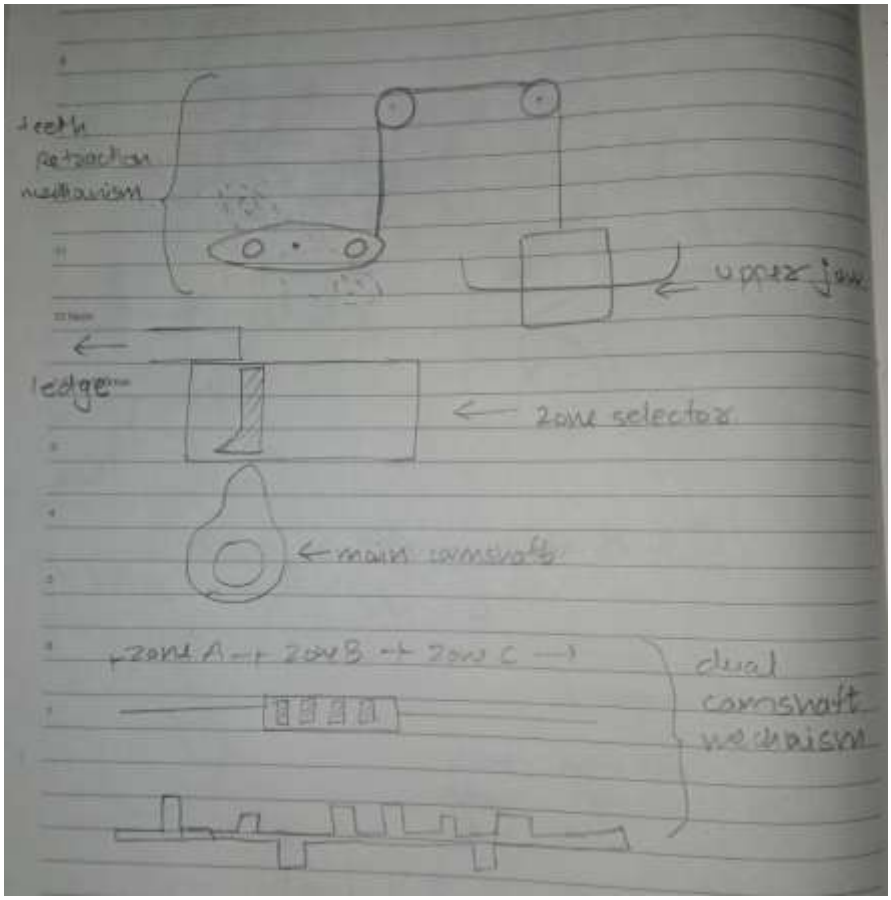


Figure 4.4: Earlier mechanical mechanism reference used during concept development.

4.10 Cam-Ring and Follower Mechanism

The ring mechanism is conceptually similar to a cam-and-follower arrangement. A cam contains a profile that converts rotary motion into controlled movement of a follower. In the vault design, the circular ring provides the rotating profile, while the knob extensions act as followers.

The ring contains ramps with different lengths or slopes. As the ring rotates, each follower encounters the corresponding ramp and is displaced. That displacement is linked to the rotational movement of the knob.

The mechanism was developed as a CAD concept rather than as a fully validated manufactured assembly. Important future checks include the exact ramp geometry, friction, follower guidance, servo torque, clearances and the return mechanism required to bring the knobs back to their initial positions.

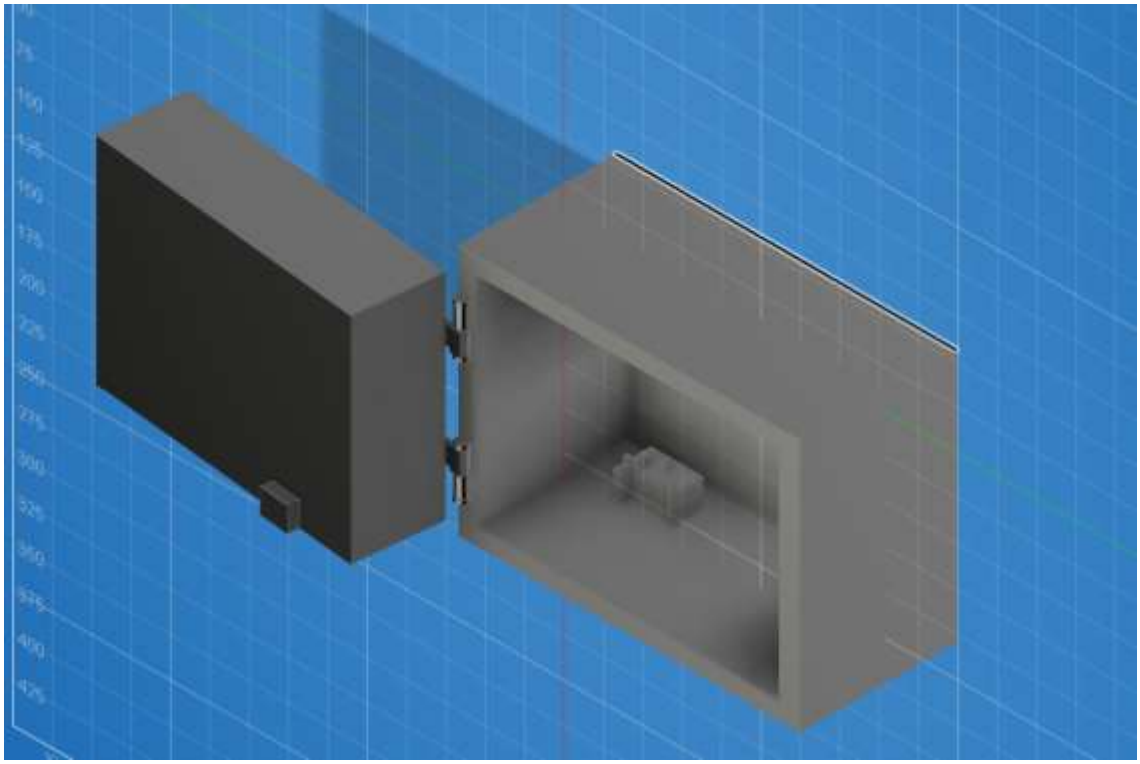


Figure 4.5: Overall vault enclosure and hinged-door CAD study.

4.11 Door, Hinge and Locking Mechanism

A separate hinges test model was created to study the main vault door. The model uses a hinged rectangular enclosure so that the front panel can swing open. The design also explores a locking component that can be moved by a servo.

An additional mechanical idea is to apply a constant outward force to the door while it is locked. When the locking element retracts, the stored force would move the door outward by a few centimetres. This would provide a clear physical indication that the vault has been unlocked and also make the door easier for the user to open.

The complete locking mechanism was not finalized during the training. The hinges test is therefore best understood as a mechanical feasibility study that informs the next iteration of the final enclosure.

4.12 Prototype Workflow and Current Status

The Hackable Vault followed an iterative development workflow. The first stage was to define the user experience: enter a password, receive HIGH/LOW feedback and physically open the vault after successful completion. The second stage translated this interaction into an electronic prototype.

The next stage involved considering how software events could produce physical movement. The early four-servo prototype demonstrated the idea directly. Later, the design was changed so that one servo could create coordinated knob movement through a cam-ring mechanism.

CAD was then used to investigate the physical geometry of the front panel and the hinged enclosure. This approach allowed the mechanical problems to be explored separately before full integration. The result is a project in which electronic and mechanical subsystems were developed in parallel.

At the end of the documented training period, the Hackable Vault was assessed as approximately 70% complete. The electronic components had been purchased, the main connections had been planned and simulated, and the principal front-panel mechanism had been designed in CAD.

The software was also substantially planned, but the complete final code had not been written. Similarly, the full vault enclosure and final door mechanism had not been completely finalized. The front-panel cam-ring mechanism and the separate hinges/locking experiment represented the most advanced mechanical work.

The project status is therefore reported as a development-stage prototype rather than a fully finished product. This distinction is important for technical accuracy. The completed parts demonstrate the skills developed during training, while the unfinished parts provide a clear future development path.

At the end of the training, the project was assessed at approximately 70% completion. The electronics had been purchased and the main connections had been simulated. The front-panel cam-ring mechanism and the hinged enclosure/locking study were substantially developed in CAD. The complete final software and full mechanical integration were not finished, so the project is documented as a development-stage prototype with clear remaining work.

CHAPTER 5

RESULTS AND DISCUSSION5.1 Simulation Results

The Wokwi simulation provided a working environment for checking the basic electronic architecture of the first vault prototype. The keypad, display, LEDs, buzzer and servo outputs were connected to the Arduino and used together by the program.

The simulation demonstrated the main interaction sequence: a level is displayed, the user enters a numerical guess, the guess is submitted, and the program provides HIGH/LOW or correct feedback. A successful guess also produces servo movement and advances the level.

The result of this stage was not the final Hackable Vault hardware, but a functional electronic proof of concept. It reduced uncertainty before moving toward a more mechanically complex design.

The programming exercises produced several useful results. The debouncing exercise established an understanding of how physical switches should be interpreted. The Gate-In/Gate-Out project demonstrated state-based counting and output control.

The first vault program brought these ideas together. It stores the current input, compares the guess against a generated password, produces conditional feedback and controls physical outputs. The use of separate functions for starting a level and handling the win state also improved the organization of the program.

The final Hackable Vault software was not fully implemented during the training period. The working first-iteration code is therefore included as evidence of the electronic game logic, while the timer, final two-OLED interface and final two-servo sequence remain development work. This distinction is maintained throughout the report so that the prototype is not presented as more complete than it was.

5.2 CAD and 3D Printing Results

The CAD phase produced several completed or substantially developed models, including phone-holder designs, a piggy bank, a piston cup and vault-related mechanical components. These models demonstrated increasing complexity as the training progressed.

The piston cup provided the clearest physical result because it was successfully 3D printed. It demonstrated the complete workflow from digital modelling to a tangible object.

The vault CAD work produced a developed front-panel mechanism and a separate hinged enclosure study. These results showed that the training had moved beyond basic modelling into mechanical design involving movement, interaction between parts and actuator placement.

5.3 Vault Design Results

The main design result of the Hackable Vault was the integration of an electronic game concept with a mechanical physical output. The project established a clear relationship between a successful password guess and a physical unlocking action.

The four-knob front-panel mechanism demonstrated a method for producing multiple visible movements using one servo. The hinges test demonstrated a separate approach for creating the actual door movement and lock release.

Although the complete physical prototype was not finished, the design work resolved several early conceptual questions and produced a basis for continued development.

5.4 Challenges and Design Modifications

The largest challenge was mechanical complexity. The early Syblin concept required many moving teeth and introduced the possibility of needing a large number of servos. A dual-camshaft and zone-selection mechanism was considered to reduce the number of actuators, but it also introduced substantial fabrication and alignment requirements.

Space was another constraint. A mechanism may work conceptually but still be unsuitable if the required servos, cams, followers and return components cannot fit inside the available enclosure.

The project also had to balance visual appeal with buildability. The design was intended to be playful and physically satisfying, but every additional moving feature increased the number of mechanical interactions that needed to be designed and tested.

The most significant design modification was the move from the Syblin monster concept to the Hackable Vault. The original concept had a strong visual identity but required a complicated tooth-retraction system.

The vault concept retained the most important interaction: the user enters guesses and the correct solution causes a physical reward. Instead of many individually moving teeth, the final design uses four coordinated knobs and a separate door mechanism.

This change reduced the number of actuators and separated the visual unlocking animation from the actual door-opening function. It also made the project more suitable for a student-scale prototype.

5.5 Validation Plan

The available results show that the project successfully progressed through concept design, electronic simulation and mechanical modelling. However, complete validation of the final system would require physical assembly and repeated testing.

Future validation should test the keypad response, display behaviour, password comparison, timer operation, servo movement, knob synchronization, locking and door release. Mechanical tests should also examine whether the cam-ring followers move smoothly and whether the door can open reliably after the lock is released.

A successful final prototype should be evaluated not only for whether it works once, but for repeatability. The mechanism should return to its initial state, tolerate repeated cycles and avoid excessive friction or misalignment.

Test	Expected behaviour	Result to record
Keypad input	Digits are accepted correctly	Pass / Fail
Wrong guess	HIGH or LOW feedback appears	Pass / Fail
Correct guess	Correct indication and level change	Pass / Fail
Timer	Countdown operates and timeout resets level	Pass / Fail
Front knobs	Four knobs rotate together	Pass / Fail
Door lock	Lock releases after successful completion	Pass / Fail
Door opening	Panel moves outward / opens	Pass / Fail
Reset	System returns to initial state	Pass / Fail

Test	Expected behaviour
Keypad input	Digits accepted and cleared correctly
Wrong guess	HIGH/LOW feedback and retry
Correct guess	Correct indication and level progression
Timer	Countdown and timeout reset
Front knobs	Four knobs rotate together
Door/lock	Lock releases and door opens
Reset	System returns to initial state

Table 5.1: Functional validation plan for the completed prototype.

CHAPTER 6

LEARNING OUTCOMES6.1 Technical Learning

The training strengthened practical understanding of Arduino-based embedded systems. Instead of only studying input/output concepts theoretically, the trainee worked with actual modules and wrote programs that interacted with physical components.

Programming learning included handling user input, variables, conditions, functions and state changes. The vault prototype showed how several small programming concepts can be combined into a complete interaction sequence.

The CAD phase added a different technical skill set. It required thinking about dimensions, part relationships, movement and assembly. The 3D-printing activity then showed how a digital model becomes a physical object.

6.2 Mechanical-Electronic Integration

One of the most useful outcomes of the training was understanding that electronic and mechanical design cannot always be treated separately. A servo may provide the required movement electronically, but the mechanical mechanism determines whether that movement reaches the intended part.

The Hackable Vault demonstrated this directly. The software can decide that a level has been completed, but the user experience depends on how that decision is translated into the rotation of the knobs and opening of the door.

Working on the front panel and hinges models therefore provided practical experience in system integration. The design had to consider servo placement, available space, movement range and the interaction between parts.

6.3 Problem Solving and Design Iteration

The training showed that the first idea is not always the most practical implementation. The project began with an Art Thief-inspired mechanism, expanded into the Syblin concept and then changed into the Hackable Vault after the mechanical complexity became too high.

This process improved problem-solving because the design was evaluated in terms of feasibility rather than only appearance. The reduction from many teeth and servos to four coordinated knobs was an example of simplifying the mechanism while keeping the main user experience.

The experience also reinforced the value of simulation and smaller prototypes. Testing an idea at a smaller scale makes it easier to identify problems before committing to a complete physical build.

6.4 Professional and Practical Skills

The training also developed practical working habits. The trainee had to discuss ideas with mentors, work alongside other students, test designs and make changes when an approach did not appear practical.

Documentation became important because the project involved several versions and different approaches. Keeping track of the earlier vault, Syblin concept, CAD files and mechanical experiments makes the development process easier to understand and continue.

The training also provided experience in presenting technical work as a combination of problem, approach, implementation and result. This is useful for future engineering projects where the final outcome must be explained to people who were not involved in the development.

CHAPTER 7

CONCLUSION7.1 Conclusion

The industrial training at Tobotics Pvt. Ltd. provided practical exposure to a combination of electronics, embedded programming, CAD modelling, 3D printing and mechatronic prototyping. The training moved from basic Arduino exercises to larger application-oriented work and finally to an integrated project involving both software and mechanical design.

The smaller exercises were important stepping stones. The debouncing program introduced reliable input handling, the Gate-In/Gate-Out project demonstrated state-based automation, and the initial vault prototype combined keypad input, password logic, feedback and servo control.

The Hackable Vault brought these skills together. Although the final system was not completely finished, the project reached a substantially developed design stage, with electronic simulation, component selection, front-panel CAD and a separate door/hinge study completed or substantially developed. The experience demonstrated the value of iterative engineering design and practical problem solving.

7.2 Future Scope

The immediate future work is to complete the final Arduino software. This includes integrating the keypad, two OLED displays, buzzer and LEDs with the two-servo mechanism, implementing the timer and adding the intended level-difficulty progression.

The mechanical work should then focus on finalizing the vault enclosure, completing the locking mechanism and validating the cam-ring mechanism with physical parts. Servo placement, follower alignment, friction, clearances and return motion should be tested repeatedly.

Once these parts are stable, the complete system can be assembled into a single prototype and tested through repeated game cycles. Additional improvements could include a cleaner enclosure, better cable management, stronger mechanical parts and a more polished user interface.

COST OF PROJECT

The cost information below is based on the approximate figures recorded during the training context. The electronic components purchased for the project were reported to cost approximately ₹2,000–₹2,500. A separate approximate amount of ₹2,000–₹2,500 was also reported for supporting items and other components. Because the available record does not provide individual bills for every item, the table should be treated as an approximate training-project cost summary rather than an audited bill of materials.

Table C.1: Approximate project cost recorded during training.

Cost category	Approximate amount
Electronic components purchased	₹2,000 – ₹2,500
Supporting / additional components	₹2,000 – ₹2,500
Approximate recorded project expenditure	₹4,000 – ₹5,000*

*The available training note describes two approximate cost groups; individual bills were not supplied. The combined figure is therefore an approximate range and should be replaced by the actual bill total if available.

APPENDIX – I

The following appendix contains the Arduino source code used for the first vault prototype. It is included as development evidence. The code demonstrates keypad input, password generation, HIGH/LOW feedback, level progression, LED and buzzer feedback, and servo movement. It is not presented as the complete final Hackable Vault software.

```
#include <Arduino.h>#include <Wire.h>#include <LiquidCrystal_I2C.h>#include <Keypad.h>#include <Servo.h>void startLevel();void winGame();constexpr byte ROWS = 4;constexpr byte COLS = 4;char keys[ROWS][COLS] = {    {'1', '2', '3', 'A'},    {'4', '5', '6', 'B'},    {'7', '8', '9', 'C'},    {'*', '0', '#', 'D'}};byte colPins[COLS] = {8, 9, 12, 13};byte rowPins[ROWS] = {2, 3, 4, 7};Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);LiquidCrystal_I2C lcd(0x27, 16, 2);Servo servos[4];constexpr int servoPins[] = {11, 10, 6, 5};constexpr int greenLed = A1;constexpr int redLed = A2;constexpr int buzzer = A3;int currentLevel = 0;int secretPassword = 00;String inputString = "";void setup(){  lcd.init();  lcd.backlight();  for (int i = 0; i < 4; i++)   {       servos[i].attach(servoPins[i]);       servos[i].write(0);   }   pinMode(greenLed, OUTPUT); pinMode(redLed, OUTPUT);    pinMode(buzzer, OUTPUT);    startLevel();}void startLevel(){    secretPassword = random(10, 99);    inputString = "";    lcd.clear();    lcd.print("Level ");    lcd.print(currentLevel + 1); lcd.setCursor(0, 1);  lcd.print("Enter Code: ");}void loop()
```

APPENDIX – I (CONTINUED)

Continuation of the prototype source code is provided on this page. The code is included as development evidence for the first vault prototype and should be read with the preceding appendix page.

```
{    char key = keypad.getKey();    if (!key)    return;    if (key >= '0' && key <= '9')    {       if (inputString.length() < 2)     {       inputString += key;       lcd.setCursor(12, 1);       lcd.print("  "); lcd.setCursor(12, 1);          lcd.print(inputString);       }   }       else if (key == '#')   {       if (inputString.length() == 0)    return;    int guess = inputString.toInt();    if (guess == secretPassword)    { servos[currentLevel].write(90);            digitalWrite(greenLed, HIGH);            tone(buzzer, 1000, 500); lcd.clear();      lcd.print("CORRECT!");      delay(1000);      digitalWrite(greenLed, LOW);      inputString = "";       currentLevel++;       if (currentLevel < 4)       startLevel();             else       winGame();     } else    {      inputString = "";       lcd.clear();       lcd.print(guess > secretPassword ? "HIGH!" : "LOW!"); digitalWrite(redLed, HIGH);       tone(buzzer, 300, 300);       delay(1000);       digitalWrite(redLed, LOW); lcd.clear();         lcd.print("Level ");         lcd.print(currentLevel + 1);         lcd.setCursor(0, 1); lcd.print("Enter Code:");     }  }  else if (key == '*')  {     inputString = "";     lcd.setCursor(12, 1); lcd.print("   ");       lcd.setCursor(12, 1);   }}void winGame(){   lcd.clear();   lcd.print("VAULT OPEN!"); lcd.setCursor(0,1);  lcd.print("*# to Reset");  bool waitingForReset = true;  while(waitingForReset)  {    char key = keypad.getKey();     if(key == '*')     {       while(keypad.getKey() != '#');       currentLevel = 0; for (int i = 0; i < 4; i++)       {           servos[i].write(0);       }       waitingForReset = false; startLevel();         }  }  }
```

APPENDIX – II

The uploaded Wokwi configuration is included as the electronic development reference for the first vault prototype. The circuit uses an Arduino Uno, 4×4 keypad, I2C LCD, four servo motors, red and green LEDs, a buzzer and supporting resistors and breadboard connections.

The final Hackable Vault is intended to use a reduced two-servo arrangement. Therefore, the Wokwi circuit should be understood as an earlier prototype and should not be confused with the final hardware architecture.

Prototype component	**Evidence / role**
Arduino Uno	Controller used in first simulated vault
4×4 keypad	Password input
I2C LCD	Level and password feedback
Four servo motors	Physical stage movement in first prototype
Red / green LEDs	Incorrect / correct indication
Buzzer	Audible status indication

APPENDIX – III

CAD AND MECHANICAL DESIGN EVIDENCE

Selected CAD evidence from the training is presented below. The figures show the front-panel mechanism, overall enclosure, piston cup and smaller CAD exercises.

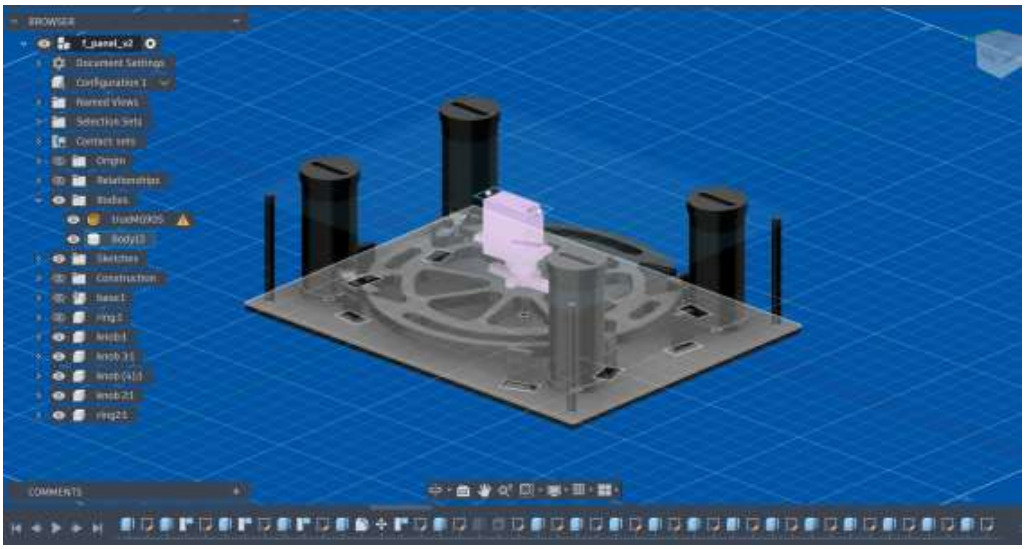


Figure A.3: Front-panel mechanism CAD evidence.

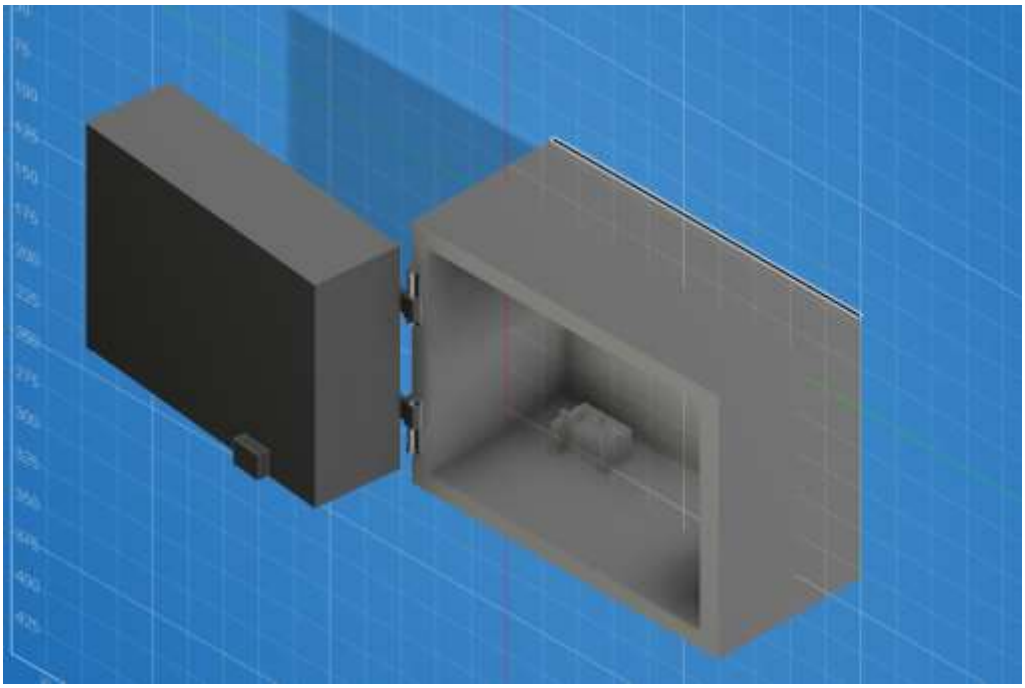


Figure A.4: Overall vault enclosure CAD evidence.



Figure A.5: Piston cup CAD evidence.



Figure A.6: Phone-holder CAD evidence.

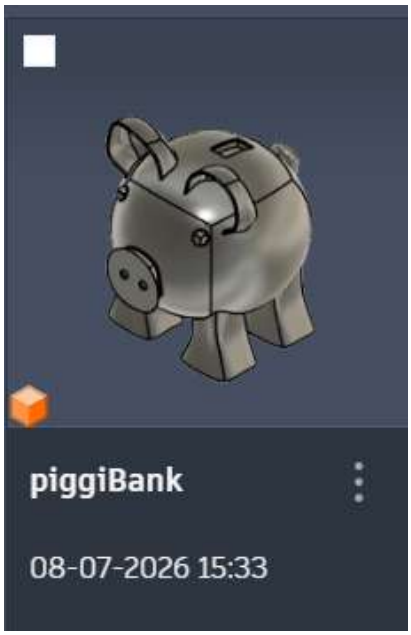


Figure A.7: Piggy bank CAD evidence.

APPENDIX – IV

TRAINING EVIDENCE

The internship certificate is included as documentary evidence of completion of the 45-day industrial training.



Figure A.8: Internship certificate supplied with the training records.

REFERENCES

[1] Jaipur Engineering College and Research Centre, “Format of Industrial Training Report, Department of Electronics & Communication Engineering, Session 2026–27,” academic year 2026–27.

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[3] Parth, “Art Thief Vault Arduino Prototype,” Arduino source code supplied as part of the industrial-training project.

[4] Parth, “Art Thief Vault Wokwi Circuit Configuration,” simulation configuration supplied as part of the industrial-training project.

[5] Parth, “Industrial Training Development Context,” training development notes covering Arduino, CAD, 3D printing and Hackable Vault development.

[6] Arduino, “Arduino Language Reference,” https://docs.arduino.cc/language-reference/.

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